

Robotics & IoT Workshop

Aishwariya Collage of Management And Research

What is a Robot?

- A robot is a programmable machine
- Robots can , think, and act



What is Robotics?

- Robotics = Electronics + Mechanical
- + Computer Science
- Includes:
 - Sensors
 - Motors
 - AI
 - Embedded systems
 - Automation



Will Break the Wall

Applications of Robotics

- Industrial automation
- Medical robots
- Military robots
- Agriculture robots
- Delivery robots
- Space exploration
- Smart homes
- Disaster rescue

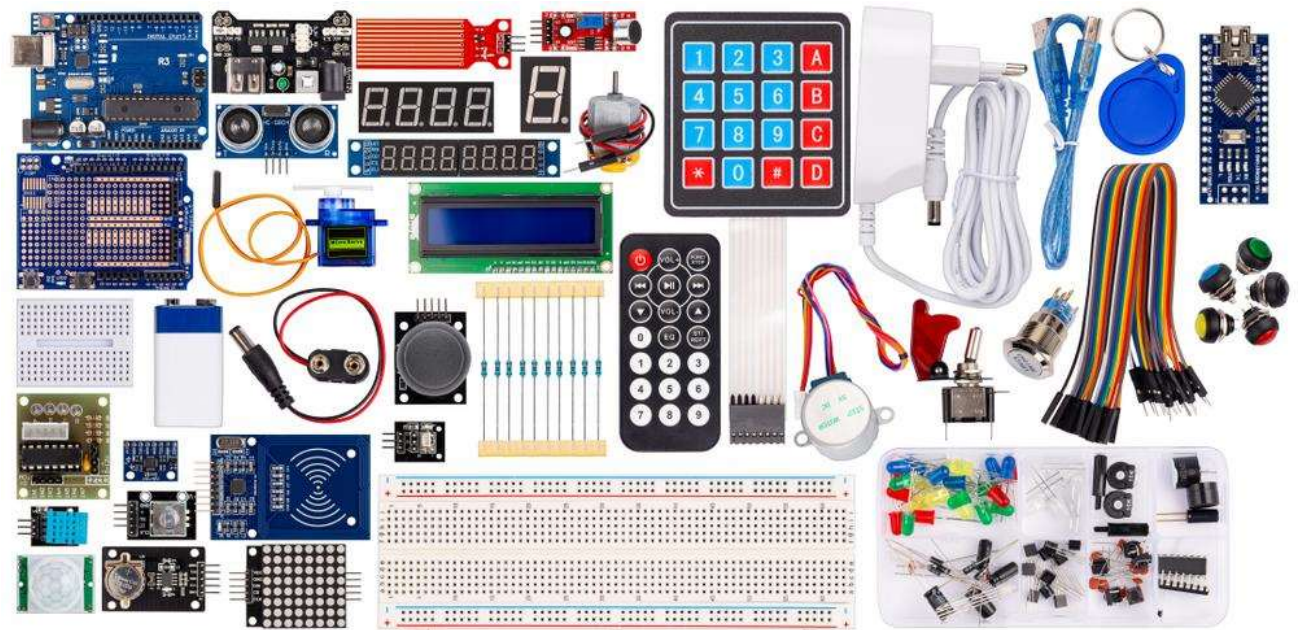
Some Experience Based Advice in Robotics

- Don't just go because it might look Fancy
- Expensive
- Time Consuming, but after all rewording

Electronics in Robotics

- Electronics are the nervous system of robots
- Sensors collect data
- Controllers process data
- Motors perform actions

Mosfet



Sensors

Internal, External

Resistive, magnetic, photo,
capacitive

Ultrasonic sensor

IR sensor

Gas sensor

Motion sensor

LDR sensor

Gyroscope

Types of Sensors



Proximity Sensor



Temperature
Sensor



Ultrasonic Sensor



Vibration Sensor



Tilt Sensor



Level Sensor



Touch Sensor



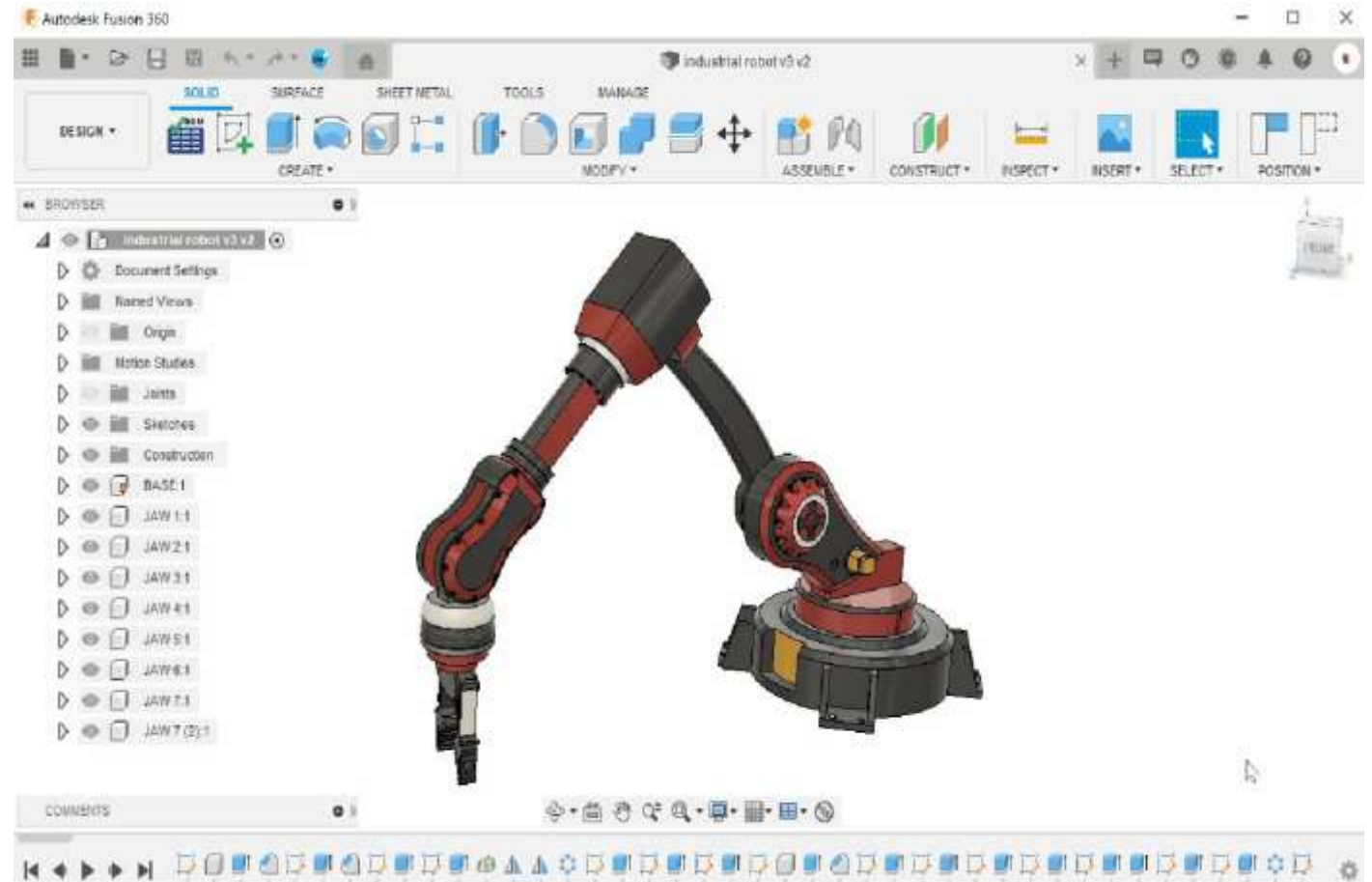
Soil moisture
Sensor



Gyroscopic
Module

Mechanical Section (Mechanical Design)

- Robot body structure
- Movement mechanisms
- Strength and balance
- Wheels, joints, arms
- Software's used
- Fusion360,
- FreeCad,
- SolidWorks,
- Catia

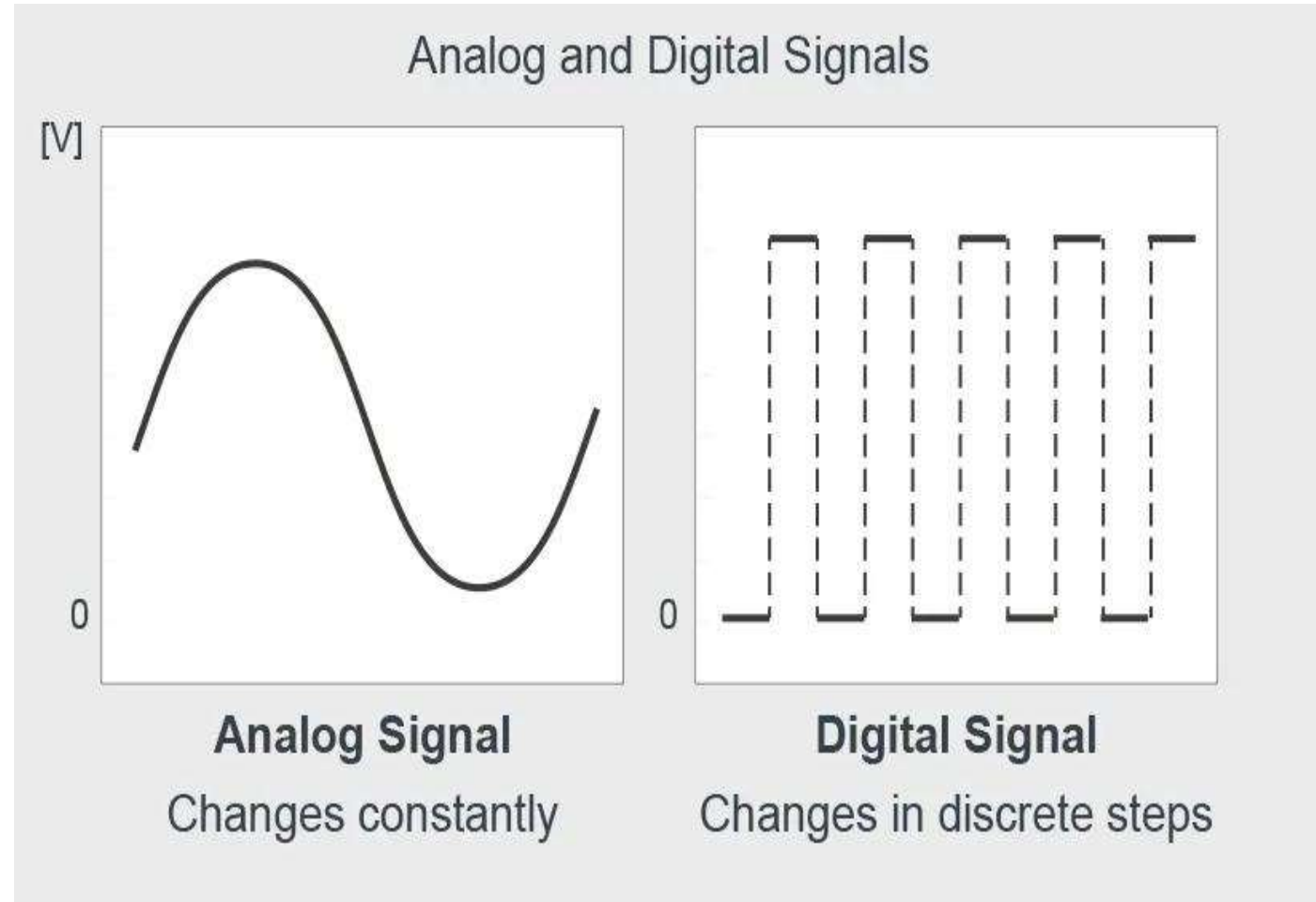


Computer Science in Robotics

- Programming controls robots
- Logic and automation
- AI and decision making
- Communication systems
- Path planning

Analog vs Digital

- **Analog**
- Continuous values
- Example: temperature sensor
- **Digital**
- ON/OFF values
- Example: button, relay
- **Example**
- Human voice = Analog
- Computer data = Digital



What is IoT?

- IoT = Internet of Things
- Devices communicate over internet
- Sensors send live data
- Remote monitoring and control
- **Applications**
- Smart lights
- Security systems
- Home automation
- Industrial monitoring
- Smart irrigation
- Vehicle tracking

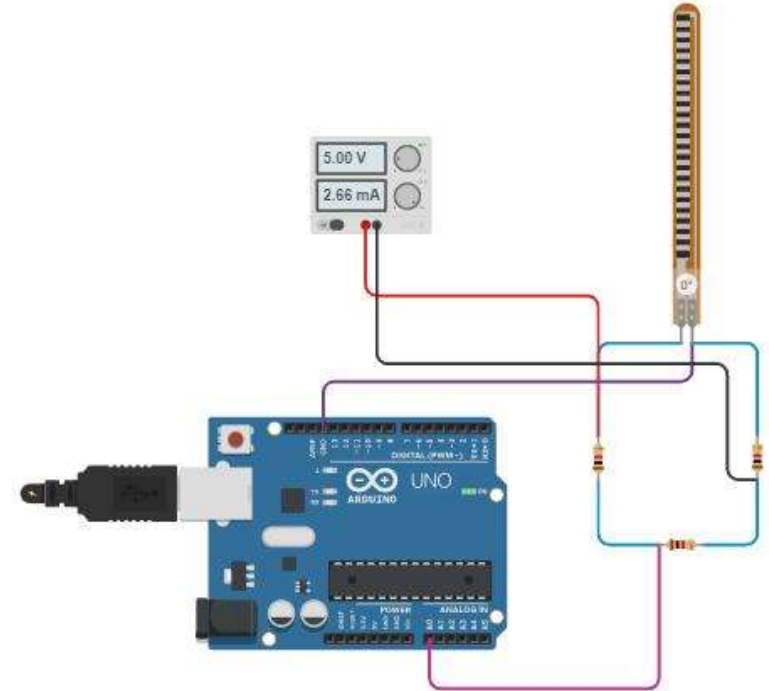
Protocols Used in IoT

- MQTT
- HTTP
- UDP
- TCP/IP
- WebSocket
- Bluetooth
- WiFi

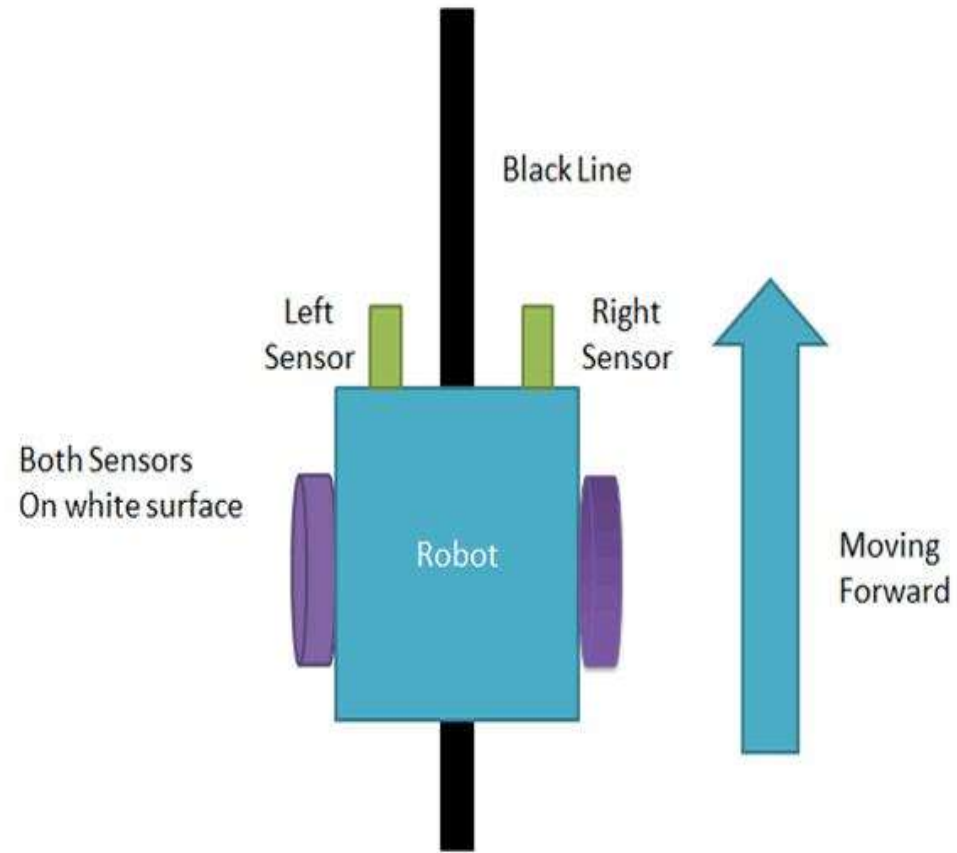
Sunday Robot Build

Tinker

- TinkerCad
- Online electronics simulator
- Test circuits without hardware
- Beginner friendly
- Great for Arduino practice



Line Follower Robot



WALL BROKEN !!

